

Practical

DSA 8302: Computational Techniques in Data Science

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Linear Programming through Excel Solver

Before moving on to excel solver, we need to have detailed question. Therefore, we start with the following problem:

A firm makes two types of furniture, chairs and tables. Profits are ₹ 20 per chair and ₹ 30 per table. Both products are processed on three machines M1, M2 and M3. The time required for each product in hours and total time available in hours per week on each machine are as follows:

Machine	Chair	Table	Available Time
M1	3	3	36
M2	5	2	50
M3	2	6	60

How should the manufacturer schedule his production to maximize profit?

STEP 1: Formulate the problem:

Let x and y be the number of chairs and tables produced per week respectively.

The LPP model is:

Maximize $Z = 20x + 30y$

Subject to constraints:

$$3x + 3y \leq 36$$

$$5x + 2y \leq 50$$

$$2x + 6y \leq 60$$

$$x, y \geq 0 \text{ (non-negativity constraint)}$$

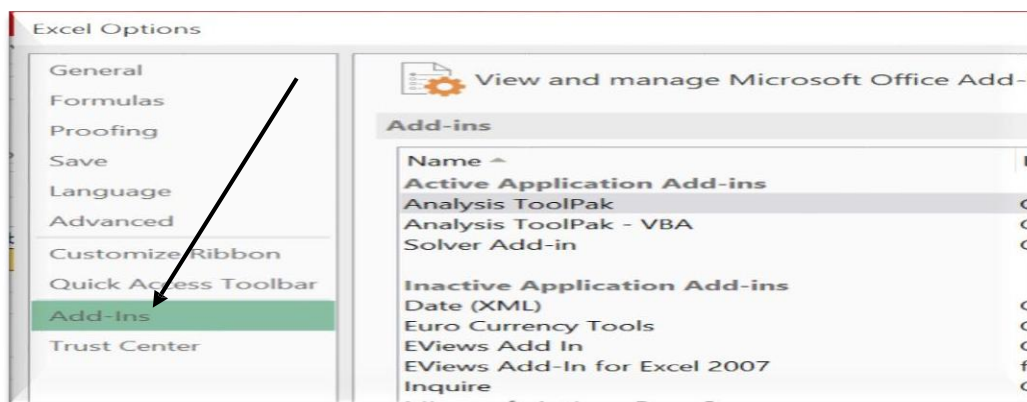
Similarly enter the formula in D6 to D8 using SUMPRODUCT formula to calculate capacity utilized of each machine

FILE		HOME		INSERT		PAGE LAYOUT		FORMULAS		DATA		REVIEW		VIEW	
 Paste		 Cut		 Copy		 Format Painter		Font 11 A⁺ A⁻ B I U   A		Alignment  Wrap Text  Merge & Center		General \$			
D6    =SUMPRODUCT(\$B\$2:\$C\$2,B6:C6)															
	A	B	C	D	E	F	G	H							
1		Chair	Table	Profit											
2	Decision variables														
3	Contribution	20	30	0											
4															
5	Constraints				Maximum Capacity										
6	Machine 1	3	3	=SUMPRODUCT(\$B\$2:\$C\$2,B6:C6)											
7	Machine 2	5	2		50										
8	Machine 3	2	6		60										
9															
10															
11															

NOTE: In the shaded cells we need to apply the formulas

STEP 4: To load the solver add-in, execute the following steps.

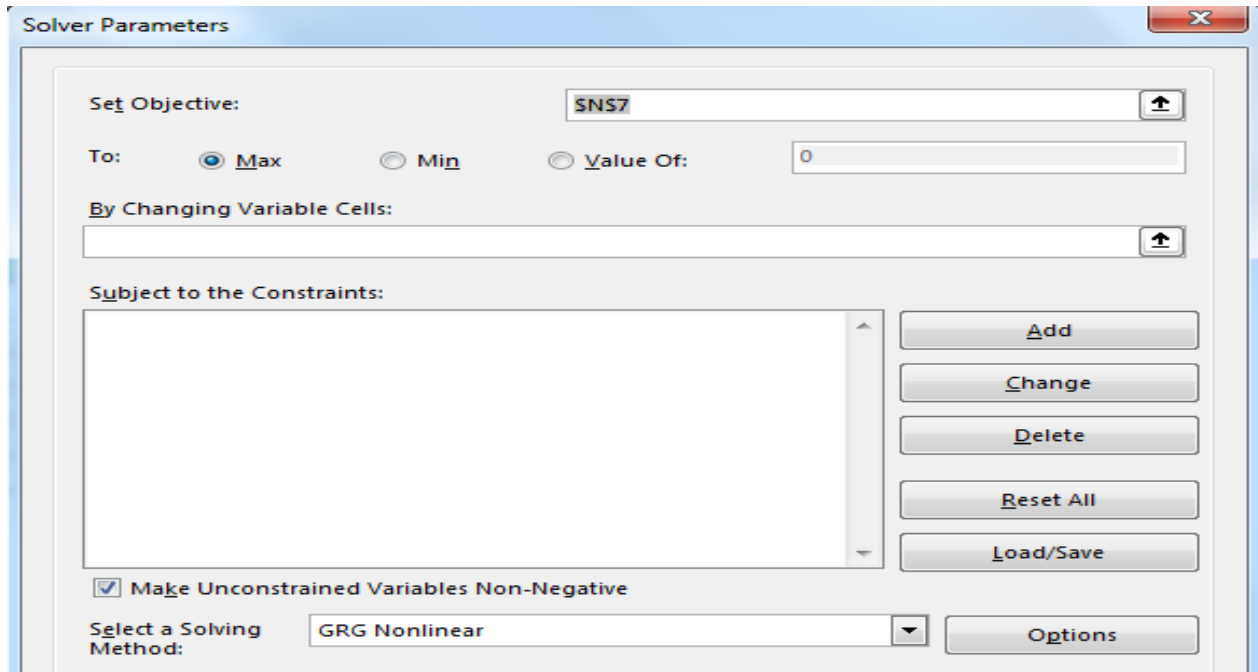
- Go to the File tab then, under File tab click on “Options”.
- Go to Add-ins under Excel Options.



- Under this select “Excel Add-ins” and click on Go.

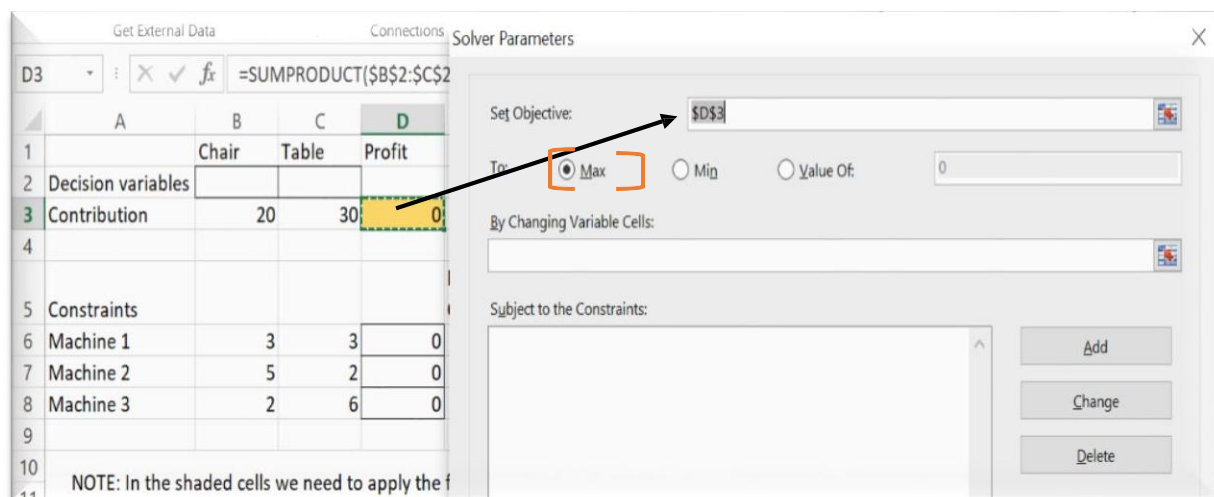
STEP 5: Solve Linear Programming through Excel Solver

To apply solver go to the DATA tab and click on “Solver” we will see below window.



The Solver Parameters dialog box is shown. The 'Set Objective:' field contains '\$N\$7'. The 'To:' section has three radio buttons: 'Max' (selected), 'Min', and 'Value Of:'. The 'By Changing Variable Cells:' field is empty. The 'Subject to the Constraints:' list is empty. The 'Make Unconstrained Variables Non-Negative' checkbox is checked. The 'Select a Solving Method:' dropdown is set to 'GRG Nonlinear'. Buttons for 'Add', 'Change', 'Delete', 'Reset All', 'Load/Save', and 'Options' are visible on the right.

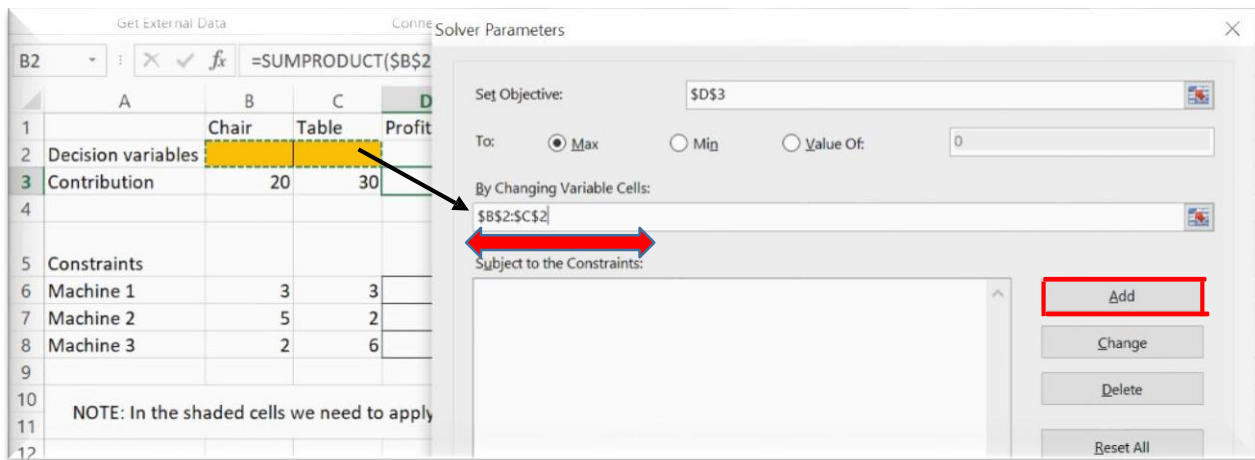
- In the above window, our first option is “Set Objective”. Our objective is to identify the “Total Profit”, so our Total Profit cell is D3, so select the cell D3 for this “Set Objective” and set it to “Max”.



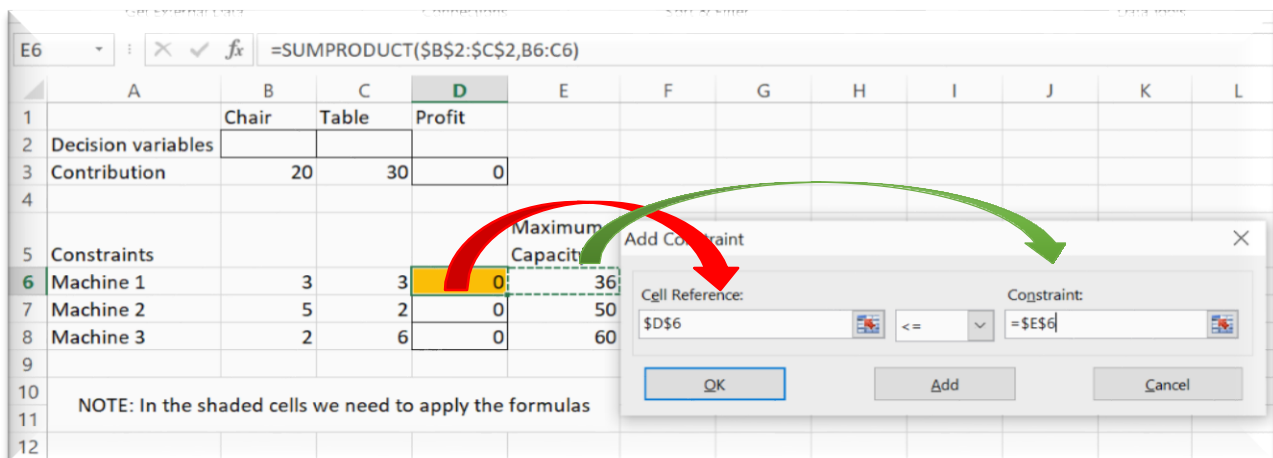
The Solver Parameters dialog box is overlaid on an Excel spreadsheet. The spreadsheet has columns A, B, C, and D. Row 1 is the header: A1 is empty, B1 is 'Chair', C1 is 'Table', and D1 is 'Profit'. Row 2 is 'Decision variables', with B2 and C2 empty. Row 3 is 'Contribution', with B3 containing '20', C3 containing '30', and D3 containing '0'. Row 4 is empty. Row 5 is 'Constraints'. Rows 6-8 are 'Machine 1', 'Machine 2', and 'Machine 3' respectively, with B6=3, C6=3, D6=0; B7=5, C7=2, D7=0; B8=2, C8=6, D8=0. A note at the bottom says 'NOTE: In the shaded cells we need to apply the f'. The Solver Parameters dialog box is open, with 'Set Objective:' set to '\$D\$3' and 'To:' set to 'Max'. An arrow points from the 'Max' radio button to the '\$D\$3' cell in the spreadsheet.

	A	B	C	D
1		Chair	Table	Profit
2	Decision variables			
3	Contribution	20	30	0
4				
5	Constraints			
6	Machine 1	3	3	0
7	Machine 2	5	2	0
8	Machine 3	2	6	0
9				
10				
11				

- Next option is “By changing variables”, in this problem, our variables are “Chairs” and “Tables”. To select a range of cell B2:C2.



- Now, the next option “subject to the constraints”, Once you click on “Add” we will see below add constraint window.



- Click on “Add” to stay back in the same window. Now in the second constraint select the value as D7 and select “<=” and under constraint select G7 cell.
Similarly, for the third constraint select the value as D8 and select “<=” and under constraint select G8 cell.

- After adding all constraints, Click on “Ok” to come out of Add Constraint window. Constraints are added as below. Now all our parameters are ready.

The screenshot shows the 'Solver Parameters' dialog box. The 'Set Objective:' field is set to '\$D\$3'. The 'To:' section has 'Max' selected. The 'By Changing Variable Cells:' field is set to '\$B\$2:\$C\$2'. The 'Subject to the Constraints:' list contains three constraints: '\$D\$6 <= \$E\$6', '\$D\$7 <= \$E\$7', and '\$D\$8 <= \$E\$8'. There are 'Add' and 'Change' buttons to the right of the constraints list.

- Now tick check box for **non-negative constraints** and select “**Simplex LP**” as solving method from the drop down. Finally, Click on **Solve**.

This screenshot shows the 'Solver Parameters' dialog box with additional settings. The 'Make Unconstrained Variables Non-Negative' checkbox is checked. The 'Select a Solving Method:' dropdown menu is set to 'Simplex LP'. The 'Solving Method' section provides instructions: 'Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.' The 'Solve' button is highlighted with a red box and a red double-headed arrow. A long black arrow points from the 'Solve' button in this screenshot to the 'Solve' button in the next screenshot.

- Finally, the answers will appear on the screen as you can see in the shaded cells below.

	A	B	C	D	E
1		Chair	Table	Profit	
2	Decision variables	3	9		
3	Contribution	20	30	330	
4					
5	Constraints				Maximum Capacity
6	Machine 1	3	3	36	36
7	Machine 2	5	2	33	50
8	Machine 3	2	6	60	60
9					
10	NOTE: In the shaded cells we need to apply the formulas				
11					
12					
13					
14					
15					
16					

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

☐ Return to Solver Parameters Dialog

☐ Outline Reports

Solver found a solution. All Constraints and optimality conditions are satisfied.

When the GRG engine is used, Solver has found at least a local optimal solution. When Simplex LP is used, this means Solver has found a global optimal solution.

Click Ok, to save the solution.

STEP 5: Write the final solution.

As per the solution given, firm can have maximum profit of ₹ 330 by producing 3 chairs and 9 tables per week.

End....